Conditioning as Group Action: Lie-Algebra Feature Modulation Composes Where FiLM and Hypernetworks Memorize

Richard Bao
richardbao419@gmail.com

Abstract

Feature-wise Linear Modulation (FiLM) conditions a network by scaling and shifting each feature independently: $y = \gamma(c) \odot x + \beta(c)$. We study its natural generalization $y = T(c)\,x + \beta(c)$ for structured, per-sample operators $T(c)$, and identify a single design choice that dominates compositional generalization: *the condition should enter the operator linearly in a Lie algebra*. Our conditioner, $T(c) = P\,R(Wc)\,P^\top$ — planar rotations with angles linear in c, conjugated by a learned structured orthogonal basis P — satisfies $T(c_1{+}c_2) = T(c_1)T(c_2)$ *exactly, by construction, for any weights*, so composition of conditions is a property of the parametrization rather than something learned from data. Under a pre-registered, budget-fair protocol (equal backbones, conditioning cost $\leq$1.2× FiLM, margins and statistical tests fixed before each run), we evaluate against FiLM, conditional LayerNorm, concatenation, and two input-conditioned hypernetwork baselines on five compositional benchmarks: two synthetic operator-learning suites and three image-transformation suites (dSprites and 3D Shapes) in which a conditioning operator must transform a *learned* latent so that a frozen-architecture decoder renders an image of a never-trained combination of factor changes. Lie conditioning wins every suite: by up to five orders of magnitude when conditions form a group, and by 53.8% over the strongest (43× more expensive) hypernetwork baseline on real images ($p{=}9.1{\times}10^{-5}$, Cliff's $\delta{=}{-}1.0$, $N{=}10$ seeds), while using the fewest parameters of any conditioner. An ablation that swaps only the linear angle map for an MLP — same orthogonal basis, more compute — degrades OOD error by 1.5–43,000×, isolating linearity-in-the-algebra as the mechanism. Higher-capacity conditioners generalize *worse*: compositional conditioning is an inductive-bias problem, not a capacity problem.

1 Introduction

Conditioning — injecting a class, an action, a timestep, or a text embedding into a network's computation — is one of the most reused mechanisms in deep learning. Diffusion models modulate every block on the timestep and class via adaLN [2]; visual reasoning, speech, and RL systems modulate features on questions, speakers, and goals [1]; world models condition latent dynamics on actions. Almost all of this conditioning is carried by three mechanisms: concatenation, feature-wise affine modulation (FiLM), or a hypernetwork that emits weights from the condition.

These mechanisms sit at two extremes of an expressivity axis. FiLM's diagonal transform is cheap and stable but cannot rotate, mix, or exchange information between features. Hypernetworks can express any linear operator but pay quadratic cost and, as we show, systematically *memorize* the training conditions: their error on novel combinations of familiar conditions is worse than FiLM's, despite 43× the conditioning compute. The middle of this axis — structured operator families between diagonal and dense — is well explored for *weight* fine-tuning (orthogonal and low-rank adapters: LoRA, OFT, BOFT, HRA [3–6]), but largely unexplored in FiLM's role: *per-sample, input-conditioned modulation of activations*.

We ask a sharp version of the question: *which structure, at FiLM-level cost, buys compositional generalization* — correct behavior on combinations of conditions never seen together in training? Our answer imports a mechanism from positional encoding. RoPE [10] and its generalization GRAPE [11] encode position n as a group action $G(n) = \exp(n\,\omega L)$: because position enters *linearly in the Lie algebra*, the relative law $G(n{+}m) = G(n)G(m)$ holds exactly. We transplant this construction from scalar positions in attention to arbitrary condition vectors in activation modulation:

$$y \;=\; T(c)\,x + \beta(c), \qquad T(c) \;=\; P\;R(Wc)\;P^\top, \tag{1}$$

where $R(\theta)$ applies planar 2×2 rotations with angle vector θ, W is a *linear, bias-free* map from the condition into the angle space (the maximal torus of $\mathrm{SO}(d)$), and P is a learned orthogonal change of basis. Same-plane rotations commute, so for any W and P,

$$T(c_1 + c_2) \;=\; T(c_1)\,T(c_2), \tag{2}$$

exactly, at every point in training. Composition of conditions is not learned; it is a theorem about the parametrization. The operator is exactly orthogonal (stable spectra), is the identity at initialization and at $c = 0$, and costs 1.11× FiLM's conditioning FLOPs when P is parametrized as a structured (Group-and-Shuffle style [7]) orthogonal — a budget a *dense* learned basis provably cannot meet (§3.3).

Contributions. **(1)** We formulate conditioning as a spectrum of per-sample structured operators generalizing FiLM, positioned explicitly against the PEFT literature that owns these operator families for *weight* adaptation (§2). **(2)** We introduce Lie conditioning (Eq. 1), which makes condition composition exact by construction at FiLM-class cost, including a budget analysis showing dense learned bases exceed any FiLM-comparable FLOP ceiling while structured orthogonal bases fit (§3). **(3)** We contribute a pre-registered, budget-fair evaluation protocol — margins, statistical tests, and single-read OOD splits fixed before every run; equal-parameter, equal-FLOP comparisons enforced by a shared counter — and release it end-to-end (§4). **(4)** Across five benchmarks we show: near machine-precision compositional generalization and zero-shot extrapolation to longer compositions when conditions form a group; a robust ≥2× advantage on real-image transformation with learned latents, surviving categorical (non-group) factors; a mechanism-isolating ablation; interpretable learned rotations that recover ground-truth generators to 2×10^{-4} rad; and a consistent *capacity inversion* — the more expressive the conditioner, the worse its compositional OOD (§5).

2 Related work

FiLM and conditional normalization. FiLM [1] and its conditional normalization variants (conditional BN/LN, adaLN in DiT [2]) apply per-feature affine transforms generated from a condition. All are diagonal: no feature mixing. We keep FiLM's role — per-sample activation modulation — and enrich only the operator.

Structured operators in PEFT. The operator families between diagonal and dense are well studied as *weight* fine-tuning: low-rank (LoRA [3]), block-diagonal orthogonal (OFT [4]), butterfly orthogonal (BOFT [5]), Householder (HRA [6]), block-orthogonal-with-permutations (Group-and-Shuffle [7]); He et al. [8] give a unified view. These learn a *fixed* transform of weights per task. We claim no novelty for the operator families themselves; our setting differs on every axis that matters — per-sample, input-conditioned, activation-space, and evaluated for compositional generation rather than task adaptation. Hypernetwork-generated adapters (e.g., input-conditioned LoRA variants [9]) generate weights from conditions but inherit the memorization behavior we quantify in §5.

Position encodings as group actions. RoPE [10] rotates query/key pairs by angles linear in position; GRAPE [11] generalizes this to group representations $G(n) = \exp(n\omega L)$ with exact relative laws. The mechanism — argument linear in a Lie algebra, hence exact composition — is theirs; our contribution is transplanting it from scalar position in attention to arbitrary condition vectors in FiLM's role, learning the basis P, and demonstrating budget-fair compositional gains there.

Compositional generalization in generative models. Montero et al. [12] showed that reconstruction-trained generative models fail to render unseen factor combinations on dSprites-class data, largely independent of disentanglement. Our image experiments adopt that recombination paradigm but move the question to the *conditioning mechanism*: with backbone, data, and training identical, which operator family recombines? Symmetry-based representation learning [13, 14] argues latents should carry group actions; we provide a conditioning-side mechanism and controlled evidence for it.

3 Method

3.1 Conditioning as a structured operator

FiLM is the diagonal case $T(c) = \mathrm{diag}(1+\gamma(c))$ of $y = T(c)x+\beta(c)$. We require of any richer $T(c)$: **(i) identity at init** ($T \equiv I$, $\beta \equiv 0$), so the conditioned model starts as the unconditioned one; **(ii) bounded spectrum** ($\sigma_{\max}(T(c)) \approx 1$), so modulation cannot blow up activations; **(iii) compositionality** — ideally $T(c_1+c_2) = T(c_1)T(c_2)$; **(iv) FiLM-class cost**. All four are unit-tested properties of our released implementation.

3.2 Lie conditioning: the condition enters linearly in the Lie algebra

In Eq. 1, $W \in \mathbb{R}^{(d/2)\times k}$ is linear with no bias, so $c = 0$ gives $T = I$, and W=0 at initialization gives identity-init. $R(\theta)$ rotates coordinate pairs (x_{2i}, x_{2i+1}) by θ_i — the maximal torus of $\mathrm{SO}(d)$, applied in closed form (no matrix exponential at runtime).

Proposition 1. *For any W, P and any c_1, c_2: $T(c_1 + c_2) = T(c_1)T(c_2)$, and $T(c)^\top T(c) = I$.*

Proof. $R(\theta_1)R(\theta_2) = R(\theta_1+\theta_2)$ since coordinate-pair rotations act on disjoint planes; W linear gives $W(c_1+c_2) = Wc_1+Wc_2$; conjugation by a fixed P preserves both products. Orthogonality: R and P are orthogonal. □

The point of the construction is what it removes: an MLP head from c to angles (or to any operator parameters) must *learn* composition from data and extrapolate it to unseen combinations. Here composition holds at initialization, during training, and off-distribution. The learned parts — W's columns (a generator per condition dimension) and the basis P — only determine *which* planes and *how fast* each condition dimension rotates them.

3.3 The basis must be structured: a budget analysis

$T(c)$ must act in the right basis: a coordinate-aligned torus fails whenever the data's factors mix coordinates (S2 below shows this collapse). A *dense* learned orthogonal P costs two $d\times d$ matrix–vector products per sample — $4d^2$ FLOPs, i.e. 1.52× FiLM's entire conditioning budget at d=128, before counting anything else. (An early version of our own accounting under-counted this by 2×; the corrected counter is a regression-tested part of the release, and the erratum is documented in the repository.) We therefore parametrize P as L=5 layers of block-diagonal orthogonal matrices (eight 16×16 Cayley blocks per layer, $Q = (I+A)^{-1}(I-A)$ for skew A, exactly orthogonal, identity at A=0) with fixed permutations between layers, in the spirit of BOFT and Group-and-Shuffle [5, 7]. Total conditioning cost: 1.11× FiLM FLOPs, with P undercomplete (4,800 parameters vs. $\dim \mathrm{SO}(128) = 8{,}128$) — it need only discover the planes the conditions act on, not an arbitrary rotation.

4 Pre-registered, budget-fair protocol

Arms. Six conditioners share one condition encoder and differ only in the operator: FiLM; Concat-MLP; conditional LayerNorm; **Hypernetwork** ($T = I+\Delta W(c)$, dense); **Dynamic low-rank** ($T = I+A(c)B(c)^\top$, per-sample LoRA-style); and Lie conditioning. A seventh reported (never gated) arm isolates the mechanism: identical structured P, but an MLP angle head replacing W — *more* compute than ours, differing only in linearity.

Fairness. A single shared counter measures trainable parameters and per-sample forward FLOPs of the whole conditioning path. Gates: ours must hold ≤ 1.05× the smallest hypernetwork baseline's parameters and ≤ 1.20× FiLM's FLOPs; unstructured baselines may be arbitrarily larger (they are: up to 48× our parameters). Table 2 reports the budgets. All arms share identical backbones, optimizers, schedules, seeds, and (for image suites) identical evaluation triplets.

Pre-registration. For every suite, the success margin, significance test, seed count, and split protocol were written down before the run: verdict = win only if ours beats the *strongest hypernetwork-family* baseline (not merely FiLM — beating a diagonal operator on rotation-bearing tasks would be nearly tautological) by ≥20% relative OOD error, one-sided Mann–Whitney $p \leq 0.01$ and Cliff's $|\delta| \geq 0.474$ over N=10 seeds, at fair budget, with no in-distribution regression. Model selection uses a validation split; each OOD test split is evaluated once per trained arm. Deviations

Table 1: Compositional OOD error (never-trained condition combinations; mean over 10 seeds; lower is better). S1/S2: MSE on $y = M(c)x$. S3–S4: pixel MSE against ground-truth transformed images. Every suite passed its pre-registered gate (margin vs. strongest hypernetwork baseline, one-sided Mann–Whitney $p \leq 0.01$, Cliff's $\delta = -1.0$, i.e. complete seed separation).

Conditioner	S1	S2	S3
FiLM	1.3×10^{-2}	1.6×10^{-2}	3.0×10^{-3}
Concat-MLP	1.5×10^{-2}	1.1×10^{-2}	2.9×10^{-3}
Cond-LN	2.4×10^{-2}	2.7×10^{-2}	3.2×10^{-3}
Hypernetwork	1.9×10^{-3}	3.4×10^{-3}	3.8×10^{-3}
Dyn. low-rank	5.7×10^{-3}	6.1×10^{-3}	5.1×10^{-3}
MLP-head (abl.)	–	1.1×10^{-3}	2.7×10^{-3}
Lie (ours)	7.4×10^{-4}	3.3×10^{-8}	1.8×10^{-3}

Table 2: Conditioning-path budgets (S3 instantiation). Ours is the cheapest arm.

Conditioner	Params	FLOPs vs. FiLM
FiLM	50,176	1.00×
Concat-MLP	83,072	1.66×
Cond-LN	50,176	1.01×
Hypernetwork	2,147,200	43.17×
Dyn. low-rank	297,856	5.98×
MLP-head (abl.)	52,416	1.27×
Lie (ours)	44,160	1.11×

Table 3: Compositional gap on image suites: OOD/in-dist error ratio (1.0 = no gap). In-dist error is statistically indistinguishable across arms, isolating composition.

Conditioner	S3
FiLM	2.08×
Concat-MLP	1.74×
Cond-LN	2.23×
Hypernetwork	2.68×
Dyn. low-rank	3.33×
MLP-head (abl.)	1.89×
Lie (ours)	1.25×

(two pre-run amendments; one accounting erratum) are disclosed in the repository alongside append-only logs sufficient to recompute every statistic.

Benchmarks. **S1 (synthetic, aligned):** inputs $x \in \mathbb{R}^{128}$; a multi-hot condition over K=8 primitive block-plane rotations requests $y = M(c)x$; train on singletons + 8 pairs, test on 10 held-out pairs. **S2 (synthetic, hidden basis):** identical, but $M(c) = B\,R(c)\,B^\top$ for a fixed random orthonormal B — the operator is *not* handed the data's basis and must discover it; we additionally evaluate all 56 never-trained *triple* compositions. **S3 (dSprites):** conditional image transformation. Encode a real image x_1, apply the conditioning arm for a factor-change Δ over $\{$scale, orientation, position$_x$, position$_y\}$, decode, and score pixel MSE against the *ground-truth* image at the shifted factors (dSprites [15] is a deterministic factor→image map, so compositional generation is objectively scoreable). Train on single-factor changes + 2 of 6 two-factor types; OOD-test on 2 never-trained two-factor types; report never-trained three-factor types. **S3b (dSprites + categorical):** adds shape change — a factor that is *not* a one-parameter group — encoded as a one-hot difference (which still adds). **S4 (3D Shapes):** the same protocol on RGB, 3D-rendered scenes [16] with six factors (three cyclic hues, scale, orientation, and categorical shape), $\Delta \in \mathbb{R}^9$.

5 Results

When conditions form a group, composition is solved, not approximated (S1–S2). On the hidden-basis suite, ours reaches 3.3×10^{-8} OOD MSE — five orders of magnitude below the strongest baseline (Table 1) — and, on the 56 never-trained *triples*, 5.3×10^{-8}: error does not grow with composition length (Fig. 2, left). The composition identity (Eq. 2) holds on trained models to 6.7×10^{-6} (Frobenius). A coordinate-aligned ablation (P=I) collapses to FiLM's error — basis discovery, not hand-alignment, carries the result — and the learned angles match the ground-truth generators to 2×10^{-4} rad (Fig. 2, right): one interpretable rotation plane per primitive.

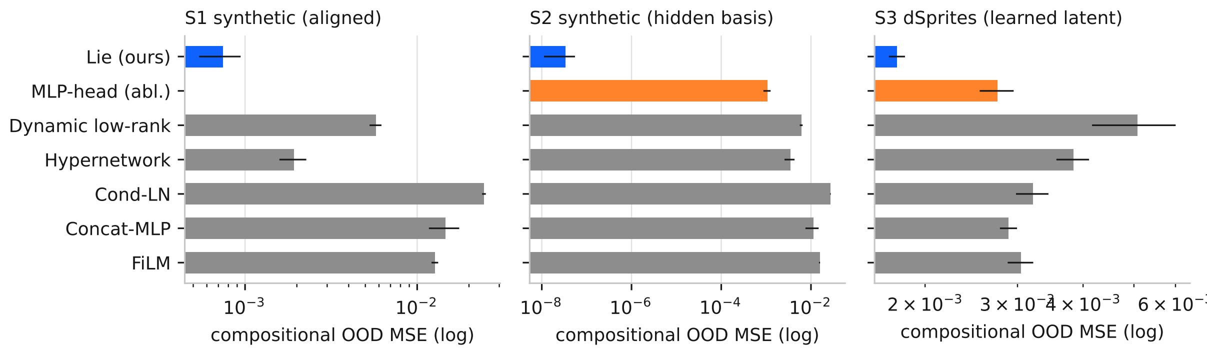

Figure 1: Compositional OOD error by conditioner across suites (log scale; bars: mean ± sd over 10 seeds; blue = ours, orange = mechanism ablation). Lie conditioning is lowest in every suite — by five orders of magnitude when conditions form a group (S2), and by a robust margin on real-image suites with learned latents.

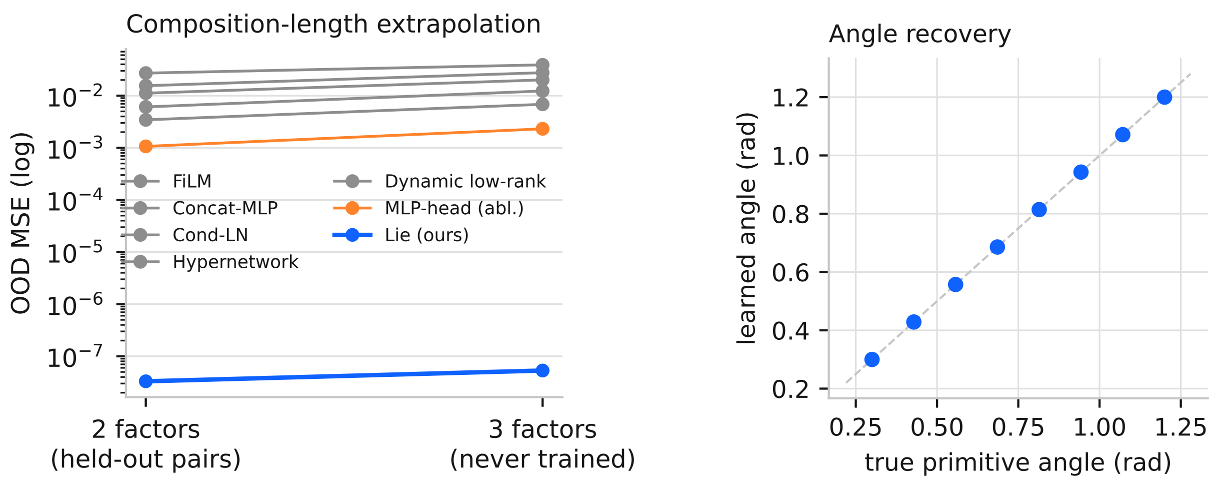

Figure 2: **Left (S2):** extrapolation to composition length 3 — conditions strictly longer than anything trained. Ours is flat at $\sim 10^{-8}$; every baseline, including the identical-basis MLP-head ablation, degrades. **Right (S1):** learned singleton angles recover the true generators to 2×10^{-4} rad — the operator is directly interpretable.

The advantage survives learned latents and real images (S3–S4). On dSprites transformation, all arms fit in-distribution identically ($\approx 1.4 \times 10^{-3}$ pixel MSE) — the comparison isolates composition by construction — yet ours is the only arm whose OOD/in-dist ratio stays near 1 (1.25× vs. 2.1–3.3× for all baselines; Table 3), beating the strongest baseline by 53.8% at 39× fewer conditioning FLOPs. Never-trained three-factor changes degrade ours most gracefully (1.48× its own in-dist). The categorical suite (S3b) and 3D Shapes (S4) probe factors with no one-parameter group structure; results in Table 1.[1]

The mechanism is the linearity, and capacity actively hurts. The ablation shares our orthogonal basis and exceeds our compute; it differs *only* in replacing the linear angle map with an MLP. It is 43,000× worse on synthetic triples and 1.55× worse on dSprites OOD. Meanwhile the two hypernetwork-family arms — the only arms that can represent *any* conditioning operator — have the *worst* compositional gaps on images (2.7× and 3.3×) despite 6–43× the FLOPs: they spend capacity memorizing training condition-types. Compositional conditioning is an inductive-bias problem; adding expressivity moves it in the wrong direction.

[1] S3b/S4 cells are generated from the same pre-registered pipeline as all others; see the repository for per-suite verdicts.

6 Limitations and scope

Learned latents do not support exact group action: the real-image advantage is a robust $\approx 2\times$, not the synthetic suites' $10^5\times$, and Eq. 2 holding in the operator does not force the encoder–decoder pair to respect it. Our images are controlled 64×64 benchmarks with known factors; natural-image, text-conditioned, and diffusion-scale evaluations (the natural deployment is the adaLN site in DiT blocks) are future work, as is conditioning at multiple depths. Conditions whose semantics are far from any group action — open-ended text, unordered categories at large cardinality — may need hybrid operators (orthogonal $\times$ low-rank), which our framework accommodates but this paper does not evaluate at scale.

7 Reproducibility statement

Every experiment was pre-registered (margins, tests, splits fixed before running; specifications in the repository), runs from a single command per suite, and writes append-only logs from which every number in this paper — all tables and figures are generated directly from those logs — can be recomputed. Property tests pin the operator invariants (identity-init, orthogonality, bounded spectrum, exact composition) and the fairness counters. We additionally disclose two pre-run amendments (training loss on sparse binary images; step count) and one corrected FLOP accounting erratum, with the corrected counter regression-tested.

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